

true dual port ram

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Overview

- instance true dual port ram & read data
- this is ETRI W6 DAY 2 ASSIGNMENT3.

Features

Constraints

- read delay is 2 clk cycles
- addr is 0 to 99
- data input is interner register and the data is 1 to 100
- write at Port A (addra 0 to 49) and read it
- writa at Port B (addrb 50 to 99) and read it
- reset is async active low
- done_w is 1 clk pulse after WRITE
- done_r is 1 clk pulse after READ

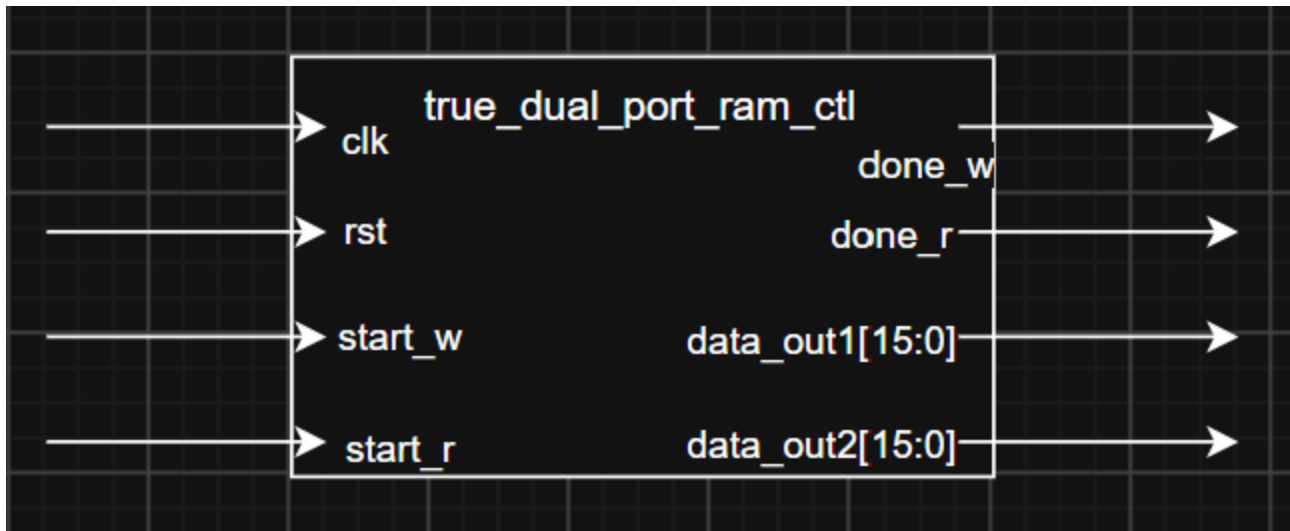
Interface

```
input logic clk
, input logic rst_n
, input logic start_w
, input logic start_r

, output logic [15:0] data_out1 // for port a
, output logic [15:0] data_out2 // for port b
, output logic done_w
, output logic done_r
```

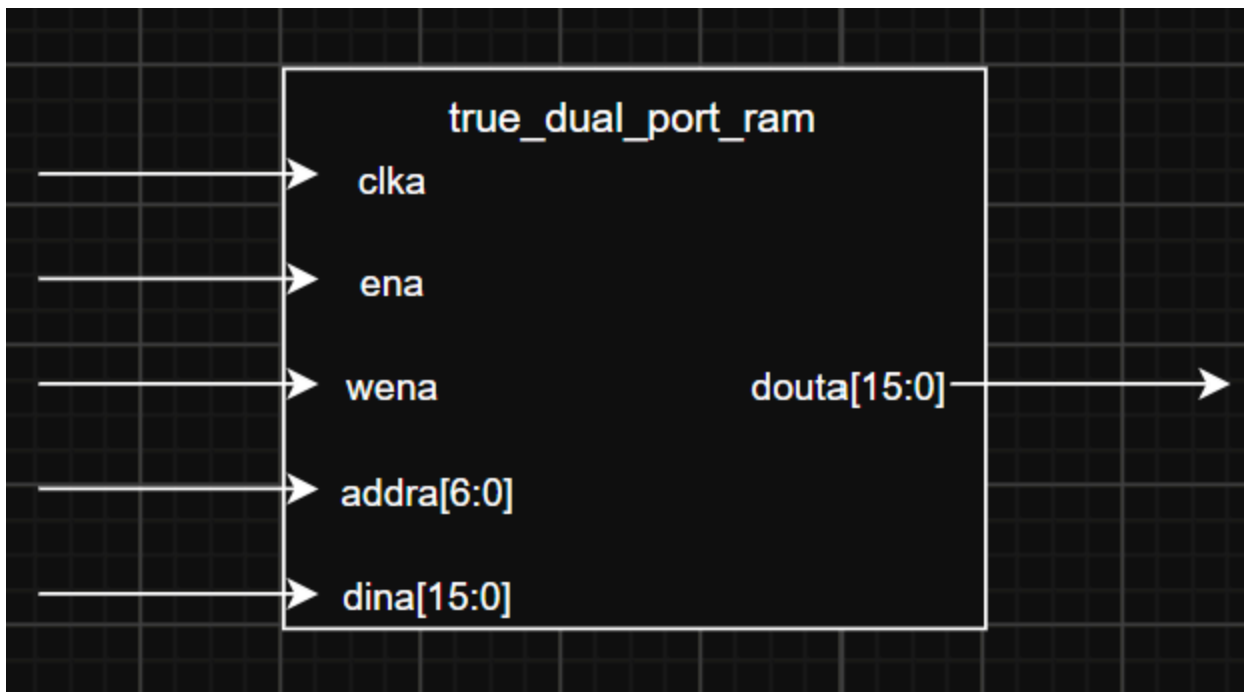
Block diagram

bd_top



s

bd_true_dual_port_ram



- port name mistake `wena` => `wea`

Registers

```
logic [6:0]      addra, addrb;  
logic [15:0]     dina, dinb;  
logic           ena, enb;  
logic           wea, web;
```

RAM if

read : addr + we + en (1clk cycle hold)=> write

write : addr + din + !we + en (2clk cycle hold)=> read

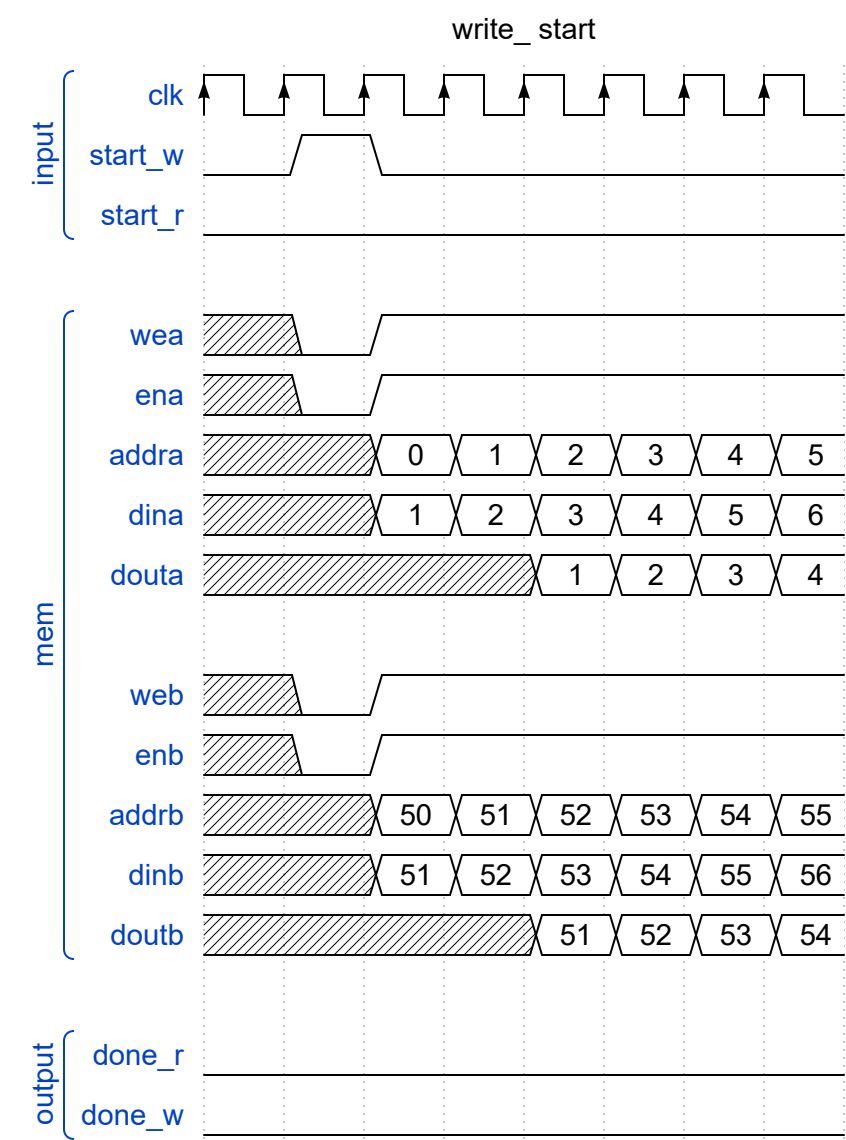
- primitives output register(so read need 2 more cycle)
- read is 1 more clk needed for en
- and then read is 1 more clk for done
- write first => dout = dina return after latency
- read first => dout = ram[addr] before write
- no_change => dout hold

this module is write first .

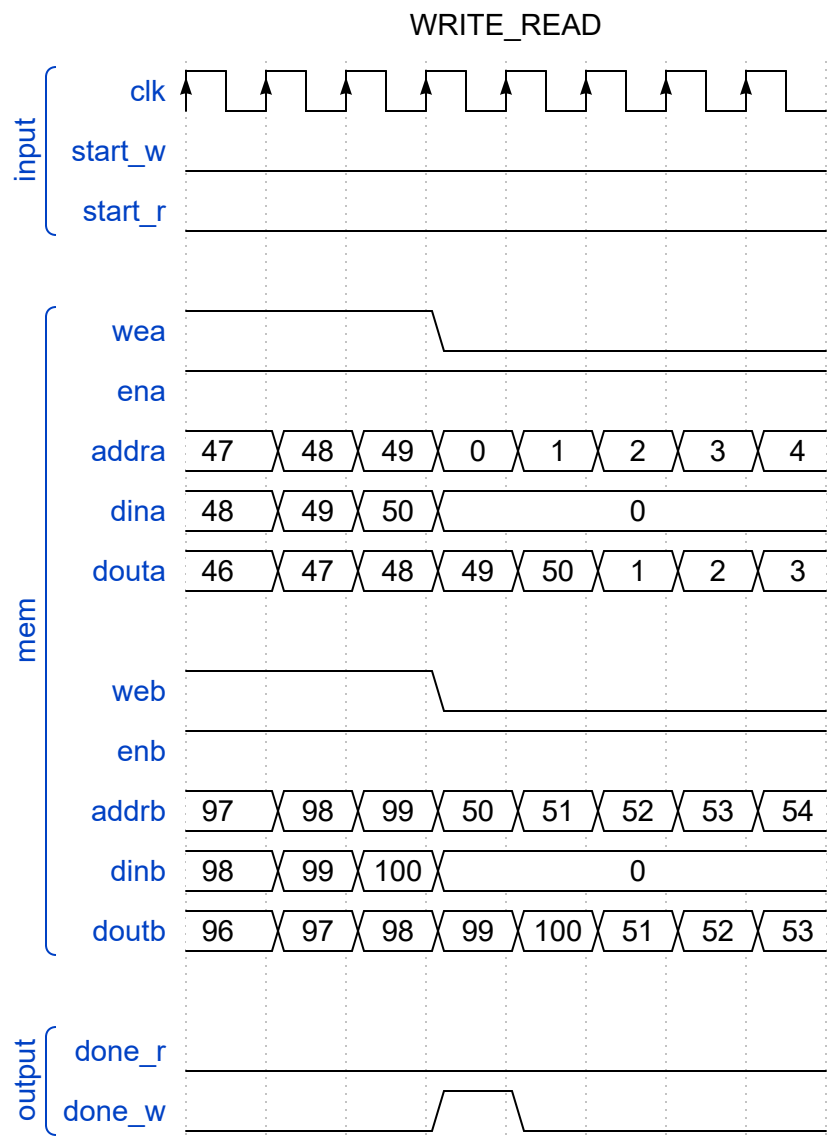
```
blk_mem_gen_0 u_blk_mem_gen_0 (  
    .clka    (clk),  
    .ena     (ena),  
    .wea     (wea),  
    .addra   (addra),  
    .dina    (dina),  
    .douta   (data_out1),  
    .clkb    (clk),  
    .enb     (enb),  
    .web     (web),  
    .addrb   (addrb),  
    .dinb    (dinb),  
    .doutb   (data_out2),  
);
```

timing diagram

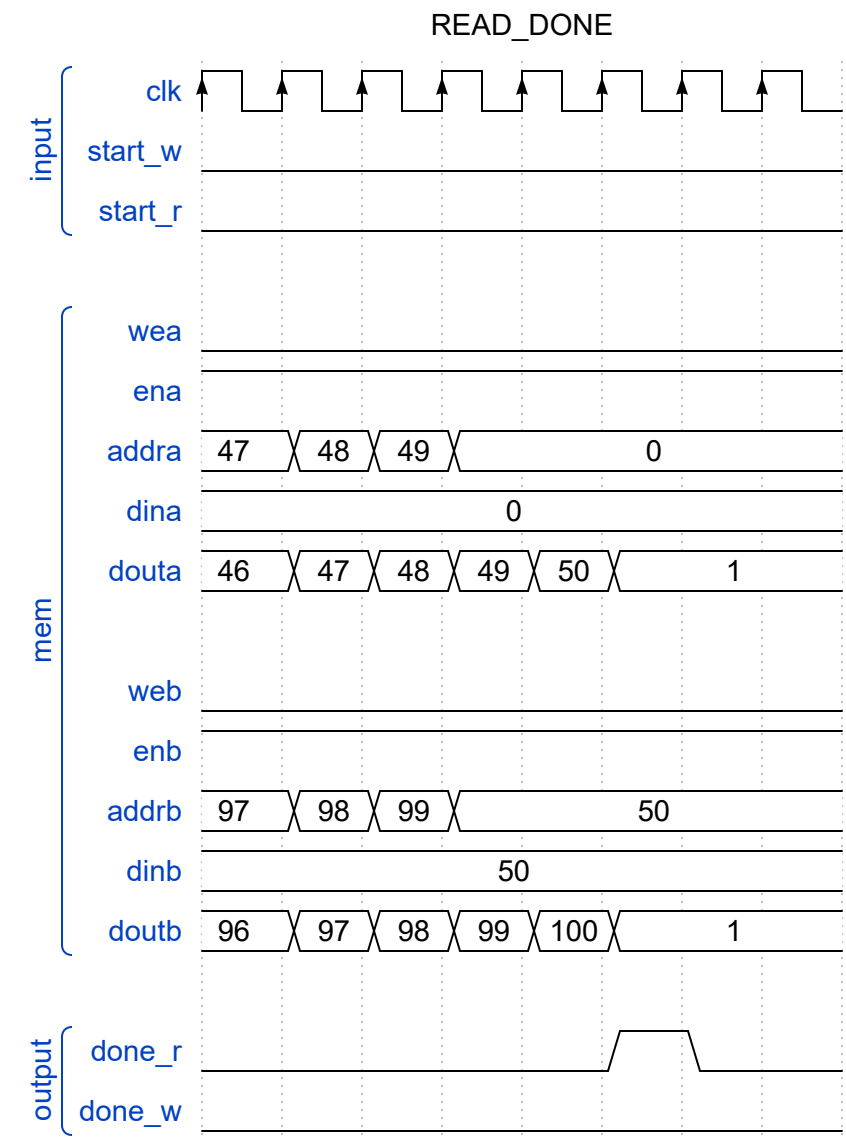
WRITE START



WRITE READ



READ DONE



state

```
write : WIDLE--start_w-->WRITE--(addra == 49)-->WDONE-->WIDLE
read  : RIDLE--start_r-->READ --(addra == 49)--> R_WAIT_DATA--1clk-->R_WAIT_DONE --1clk-->
RDONE -- 1CLK --> RIDLE
```



Syntax error in text

mermaid version 11.18.0

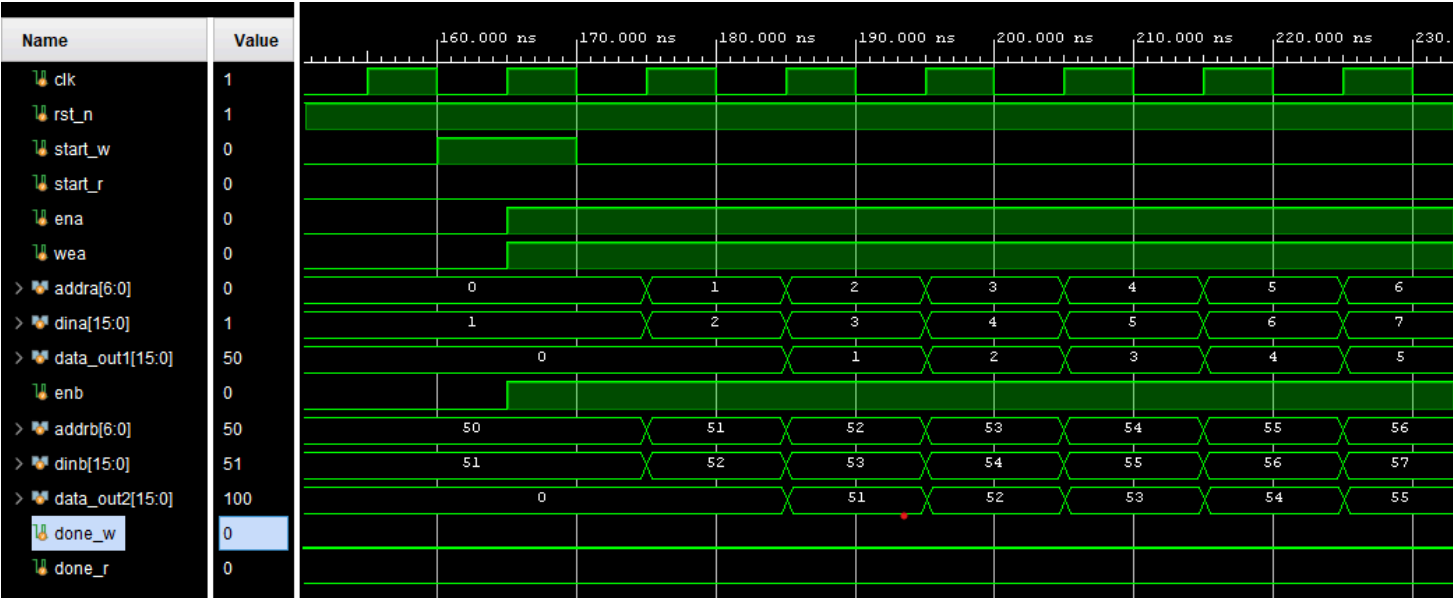
Test

Test scenario

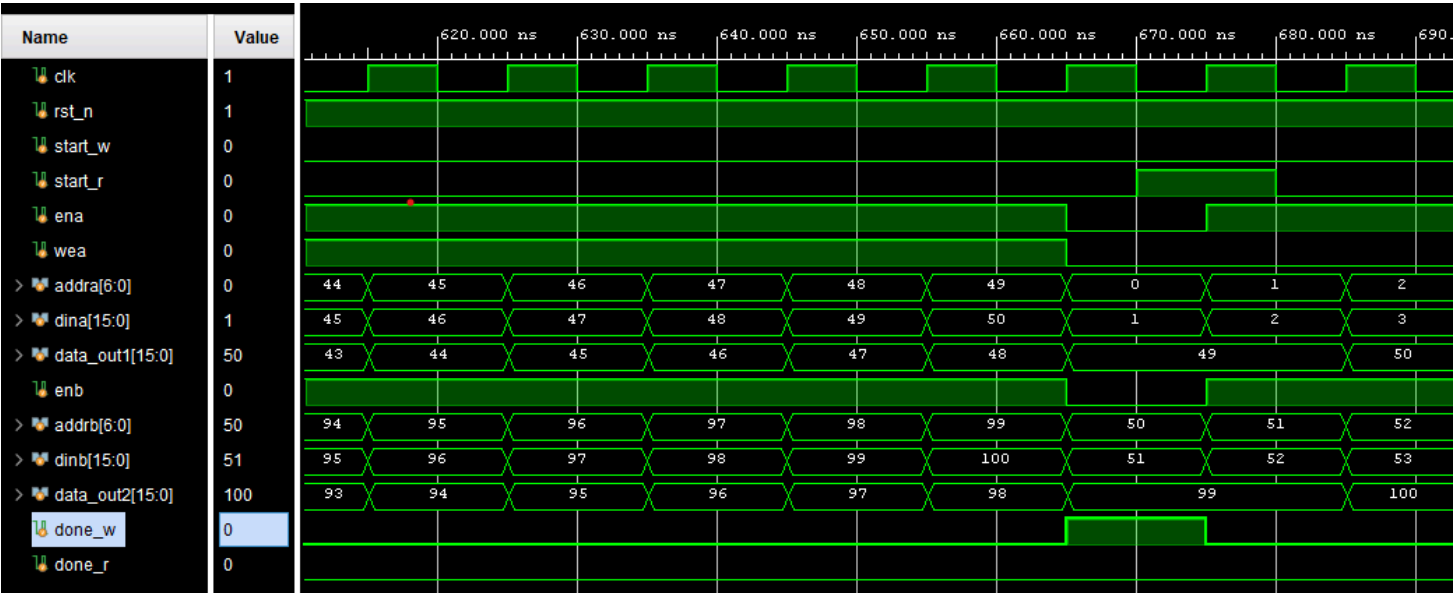
1. repeat(15) @posedge clk; // po sim때문에 10클럭 넘기고 수행
2. reset
3. start
4. wait(done)

be_sim

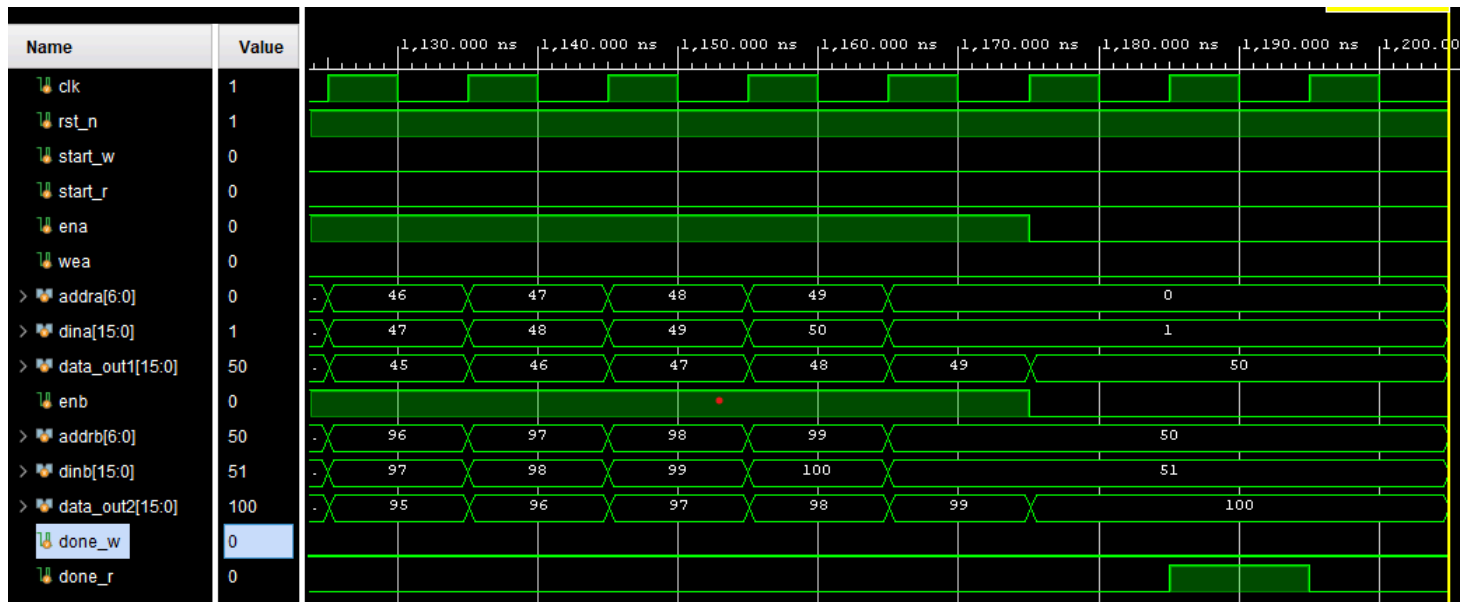
be_sim_write_start



be_sim_write_read

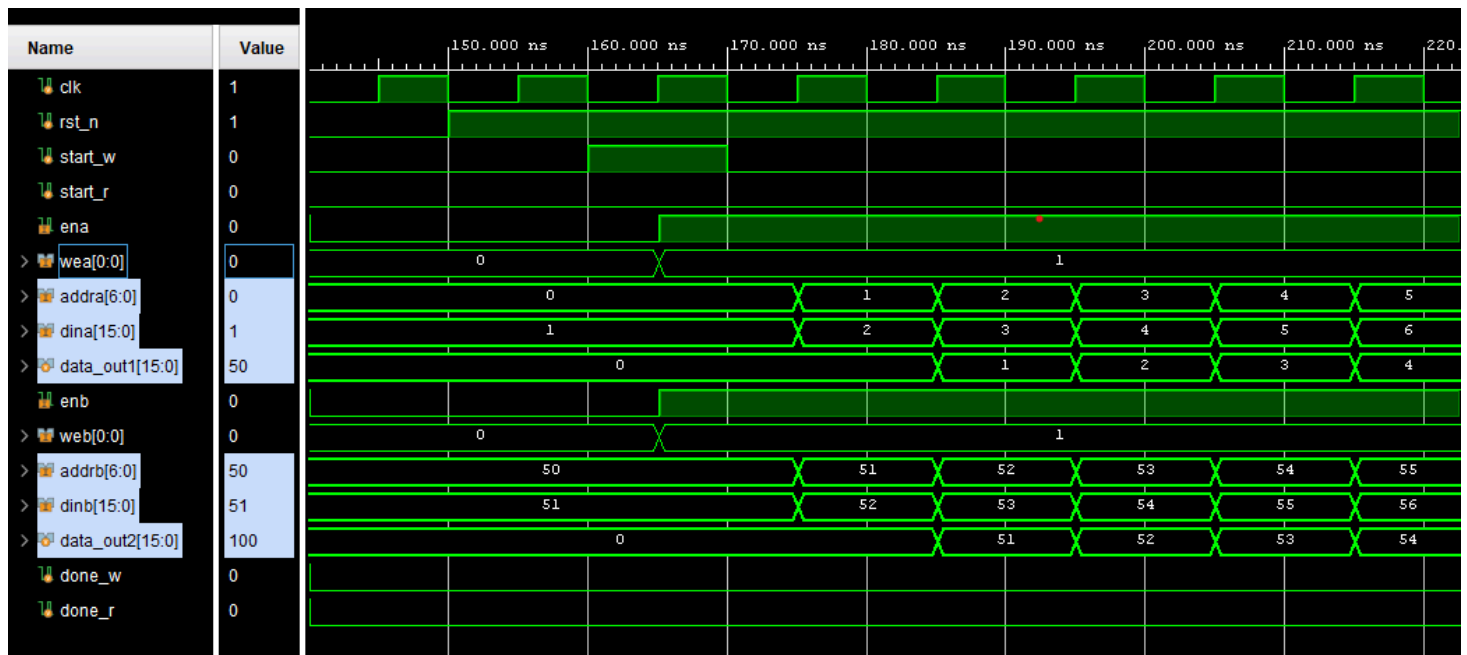


be_sim_read_done

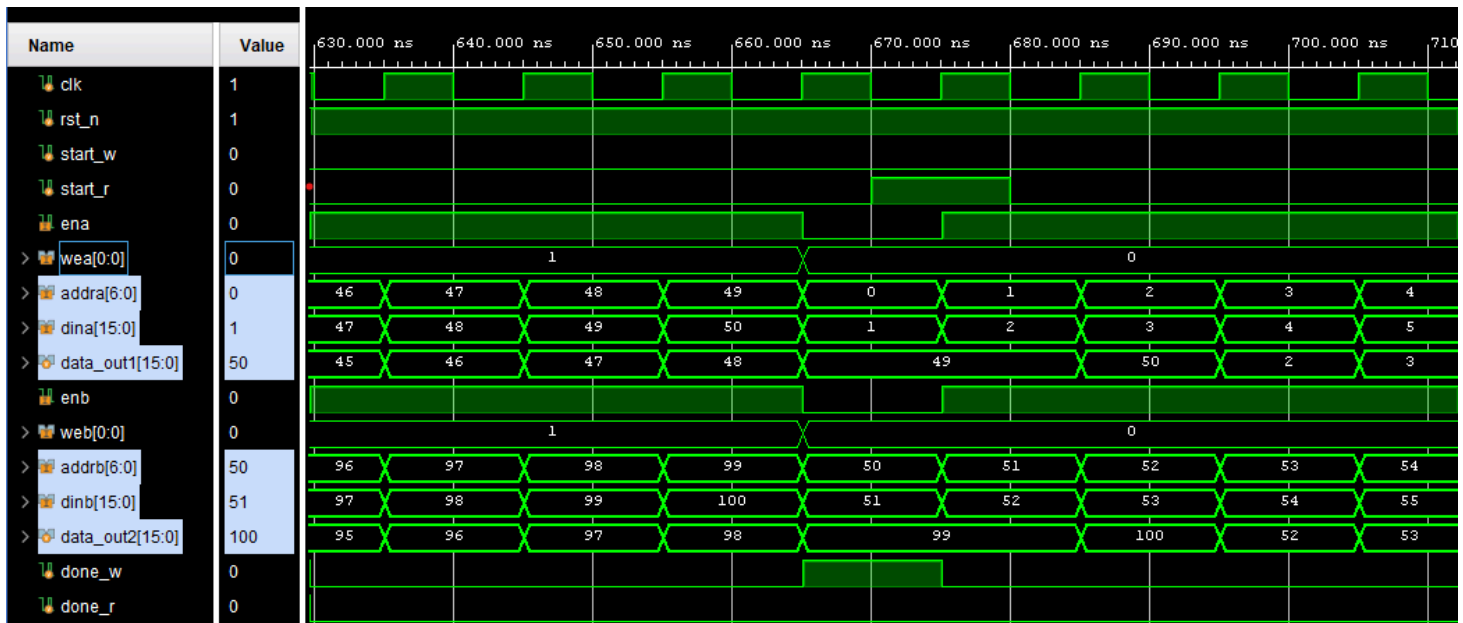


po_sim

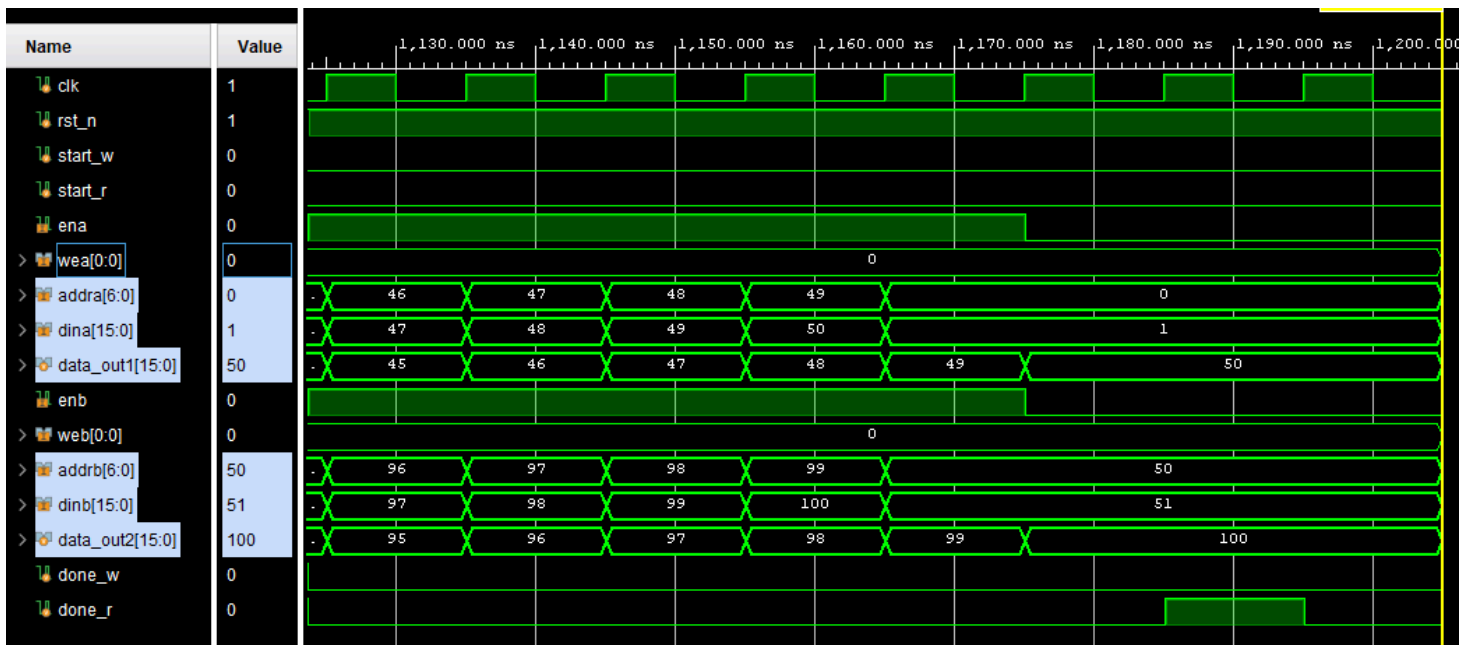
po_sim_write_start



po_sim_write_read



po_sim_read_done



Trouble shooting